

The LNM Institute of Information Technology

Title : LUSIP Report (May 2024 – July 2024)



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Date : 22 July 2024

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Project : Performance Prediction of Biomedical

Antenna using Deep Learning

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1. Introduction

Developed a Python-based application to predict the Inset Feed Length and Notch Gap for microstrip line antennas using advanced machine learning and deep learning techniques. This involved generating comprehensive training data through the analysis of various antennas, with parameters such as frequency, substrate height, return loss and permittivity being systematically varied. By integrating sophisticated machine learning and deep learning models, the application significantly improved the accuracy of predictions for antenna design parameters, thereby enhancing overall design efficiency and performance.

2. Significance

The optimization of antenna design enhances the design process for patch antennas by predicting critical parameters with improved accuracy, resulting in better performance and efficiency in wireless communication systems. This project makes a significant scientific contribution to the fields of machine learning and antenna design, offering new methodologies and approaches that can be applied to similar problems in electronics and telecommunications. By integrating advanced machine learning and deep learning models, the project not only improves the precision of antenna parameter predictions but also advances the overall design efficiency, thereby benefiting both academic research and practical applications in the industry.

2.1 Significance in Medical Field

1. **Enhanced Medical Device Communication:** Improved antenna designs can lead to better signal transmission for wireless medical devices, such as remote patient monitoring systems and wearable health trackers, ensuring accurate and reliable data collection.
2. **Enhanced Medical Imaging:** For medical imaging devices that use wireless communication, optimized antennas can improve the quality and reliability of image data transmission, contributing to better diagnostic accuracy.
3. **Telemedicine:** Reliable antennas can improve the performance of telemedicine systems, ensuring high-quality video and audio communication between patients and healthcare providers, which is crucial for remote consultations and diagnostics.

2.2 Implantable Device : Specific Improvements

1. **Pacemakers:** Improved antennas can enhance the wireless communication between a pacemaker and external monitoring systems, allowing for more reliable transmission of heart data and real-time adjustments.
2. **Neurostimulators:** Devices such as deep brain stimulators (DBS) and spinal cord stimulators benefit from optimized antennas to ensure reliable wireless communication for adjusting stimulation parameters and monitoring patient response.
3. **Cochlear Implants:** Better antenna designs in cochlear implants can improve the quality and stability of the wireless connection between the external sound processor and the internal implant, enhancing auditory experience for users.
4. **Insulin Pumps:** For insulin pumps that are implanted or worn, optimized antennas can improve the reliability of data transfer and control signals, leading to better management of diabetes..

5. Implantable Glucose Sensors: Optimized antennas can improve communication between implantable glucose sensors and external monitoring systems, providing more accurate and timely glucose level readings for diabetes management.
6. Neural Interfaces: Advanced antenna designs can enhance the functionality of neural interfaces used for brain-machine interfaces (BMIs) and other neuroprosthetics, improving communication and control.

2.3 Other Practical Application

1. Telecommunications: Optimize antenna designs for mobile networks, improving signal strength and coverage in smartphones and base stations.
2. Satellite Communication: Enhance the design of antennas used in satellites, leading to better data transmission and reception.
3. Consumer Electronics: Apply optimized antenna designs to products like smart TVs, GPS devices, and wearable technology for better connectivity.
4. Defence and Aerospace: Develop advanced antenna systems for military and aerospace applications, where precision and performance are critical.
5. Internet of Things (IoT): Design efficient antennas for IoT devices, enabling better communication between sensors, smart appliances, and networks.
6. Agriculture: Improve connectivity for agricultural sensors and drones used in precision farming, enhancing data collection and analysis.
7. Energy Sector: Design antennas for energy management systems and smart grids, facilitating efficient communication in energy distribution and monitoring.
8. Broadcasting: Optimize antennas for television and radio broadcasting to enhance signal quality and coverage.

3. Antenna Theory

3.1 Basic Characteristics of Antenna

A microstrip line antenna operates by radiating electromagnetic waves through a flat, rectangular patch of metal placed on a dielectric substrate with a ground plane on the other side. The patch, usually made of conductive material like copper, is excited by a feed line, which can be a microstrip line or a coaxial probe. When a signal is applied, it creates an electric field between the patch and the ground plane, causing the antenna to radiate. The dimensions of the patch and the properties of the substrate, such as its permittivity and height, determine the resonant frequency and bandwidth of the antenna, making it suitable for various applications in wireless communication systems.

Microstrip patch antennas are comprised of a very thin metallic strip also called patch ($t \ll \lambda_0$, where t is the thickness of copper patch, and λ_0 is the free space wavelength) placed above a ground plane at height ' h ' where ' h ' is a very small fraction ($h \ll \lambda_0$, usually ranges from 0.00320 to 0.0520). And between the strip (patch) and the ground plane placed a dielectric sheet (the substrate)

3.2 Geometrical Configuration

The top view and cross-sectional view of the microstrip patch antenna are shown in Fig. 1 and Fig. 2, respectively.

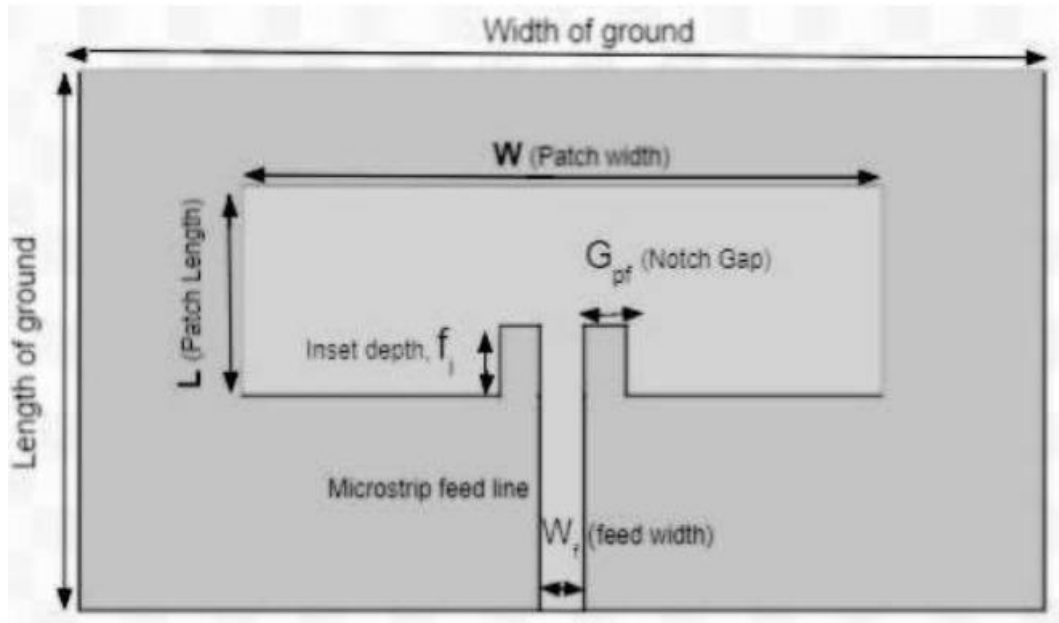


Fig. 1

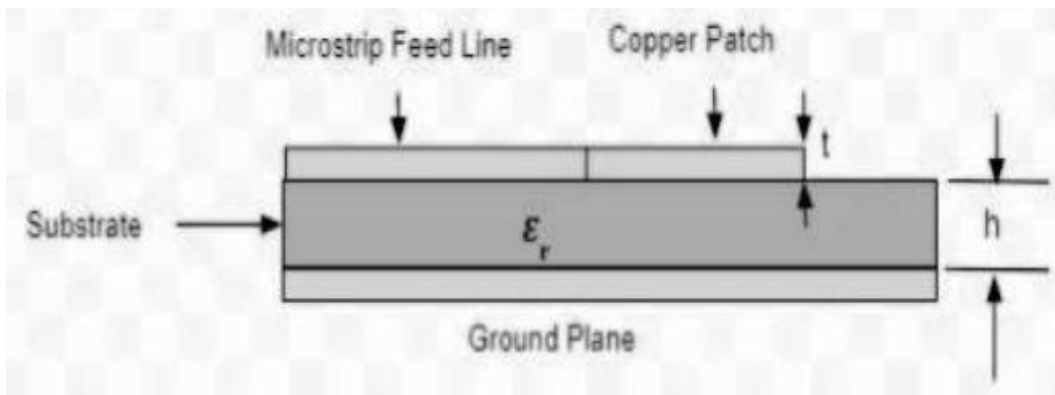


Fig. 2

The substrate dielectric constant, substrate height, patch width, patch length, inset depth, feed line width, and notch gap are denoted by ϵ_r , h , W , L , f_i , W_f , and G_{pf} , respectively.

The input impedance (Z_o) is considered constant with a value of 50Ω .

4. Training Data

4.1 Antenna Combination

Training data was obtained from HFSS analysis of the antenna for various parameter combinations

Parameters :

Substate Height(in mm) : 0.2, 0.4, 0.8, 1.6

Substrate Permittivity : 2.2, 3.35, 4.4, 9.4, 10.2

Frequency(in GHz) : 0.915, 2.45, 4, 5.8

Total 80 Antenna

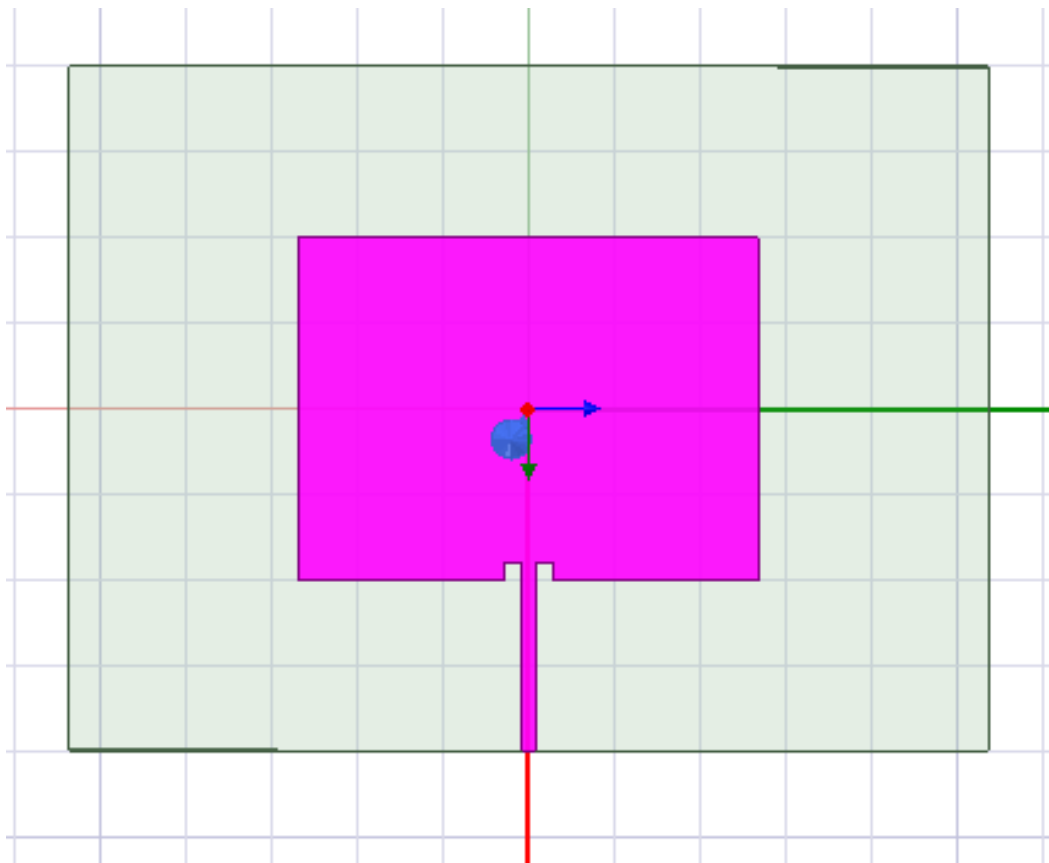


Fig. Screenshot of Patch Antenna

4.2 Sweep Setup

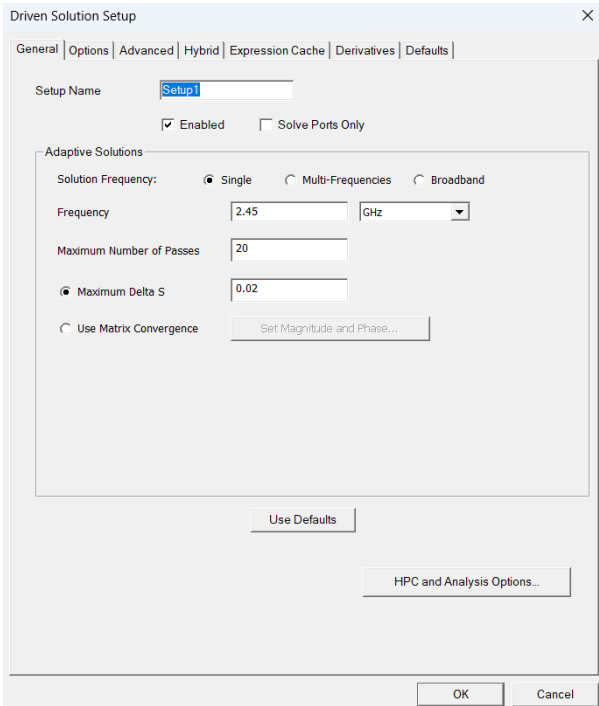


Fig. Analysis Setup

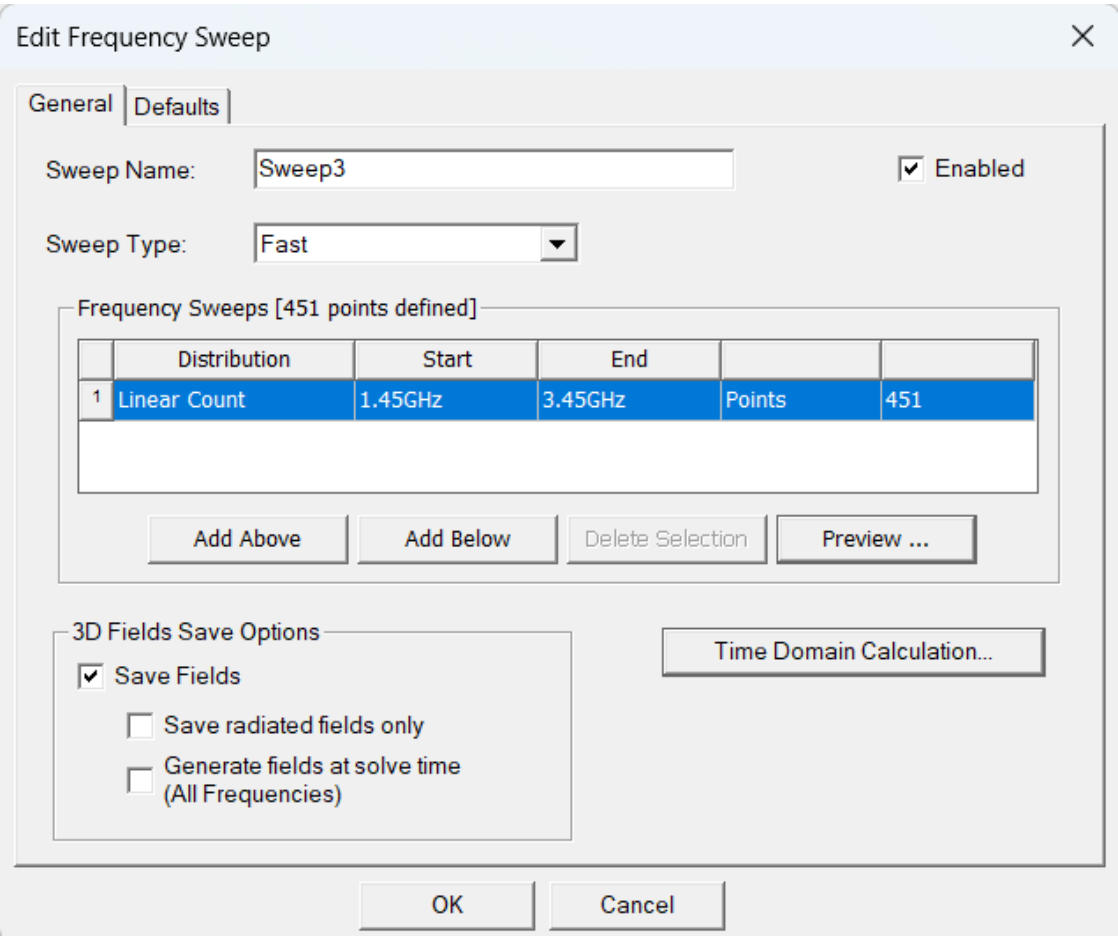


Fig. Sweep Setup

4.3 Parametric Analysis

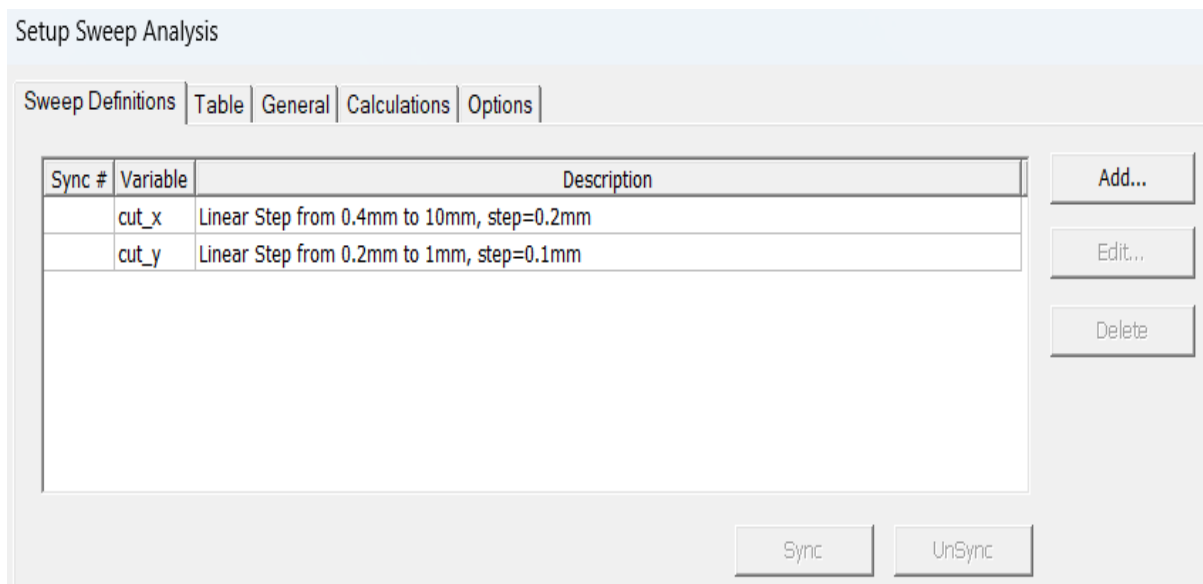


Fig. Parametric Analysis

For Parametric Analysis Feed Length(cut_x) and Notch Gap(cut_y) were varied as shown in Fig.

cut_x is varied from 0.4mm to (L/2) for a step size of 0.2mm.

cut_y is varied from 0.2mm to 1mm for a step size of 0.1mm.

4.4 Cleaning of Result Data

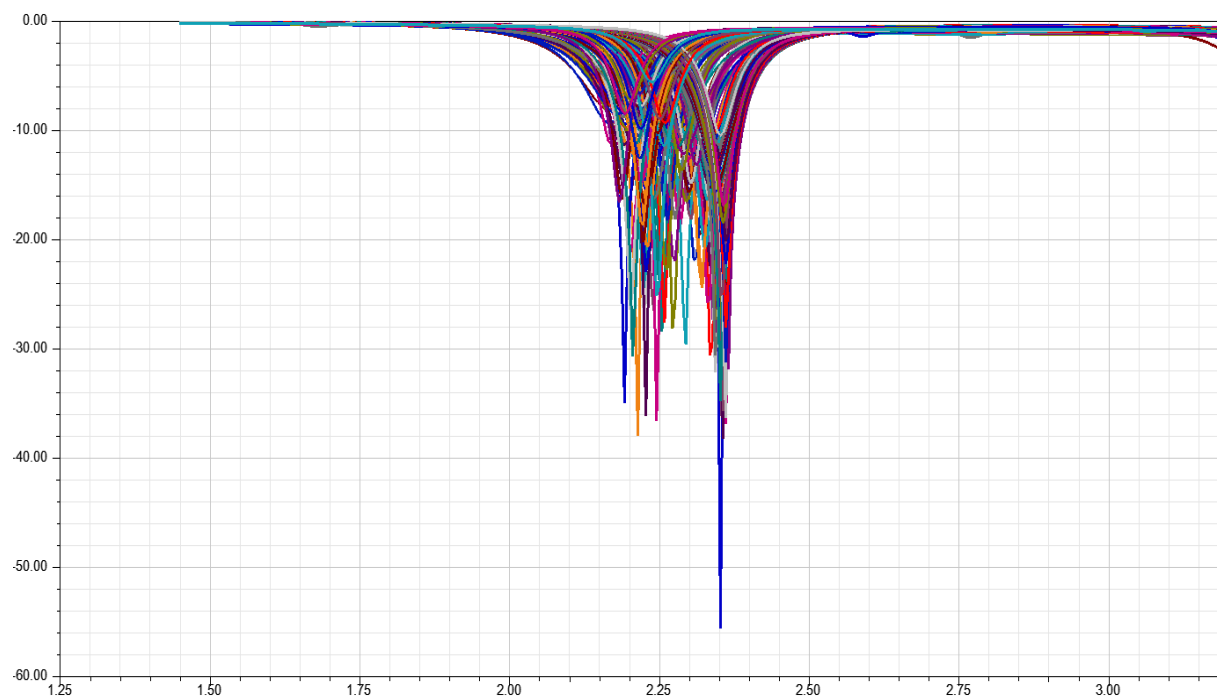


Fig. Analysis Results

Fig. represent Plot of Return Loss(S11) v/s Frequency(GHz) for each Parametric Combination This data was then imported to csv. Where using Pandas and Numpy, Minima having loss $<-20\text{Db}$ and loss $>-40\text{Db}$ lying in ± 0.3 Ghz range of Resonant Frequency were extracted in a Separate csv file.

5. Model Selection & Training

5.1 Standardization & Train Test Split

Data Standardization: Used the Standard Scaler module from the sci-kit-learn library to standardize the data through feature scaling, which accelerates model training and aids in finding the global minima.

Formula : $z = (x-u)/s$; u = Mean & s = Standard Deviation

Train-Test Split: After feature scaling, the dataset was randomly split into training and testing sets using the `train_test_split` module from the sci-kit-learn library. Here Train split ratio is 0.7, means 70% data is used for training and 30% is used for testing the model.

5.2 Model Used : FNN (ANN)

A Feedforward Neural Network (FNN) is a foundational type of artificial neural network characterized by connections between nodes that do not form cycles. It consists of an input layer, one or more hidden layers, and an output layer. Each layer is made up of nodes (or neurons), where the input layer's nodes represent the input features and the output layer's nodes represent the predicted output. The nodes in each layer are connected to every node in the next layer through weighted connections, with no connections between nodes within the same layer and no cycles or loops in the network.

In an FNN, input data is fed into the input layer, passed through the hidden layers, and finally processed by the output layer to produce a prediction, a process known as a forward pass. The network is trained using a labeled dataset, adjusting the weights of the connections based on the error of its predictions through a process called backpropagation. This involves calculating the gradient of the loss function and using optimization algorithms like gradient descent. The difference between the predicted output and the actual output is quantified using a loss function, with common examples being Mean Squared Error (MSE) for regression tasks.

The ANN model structure was developed using TensorFlow's Keras Sequential API, resulting in a Feedforward Neural Network (FNN) with a total of seven layers. The model includes an input layer with four neurons, corresponding to the four independent variables, an output layer with one neuron, and five hidden layers comprising 8, 8, 16, 32, and 64 neurons respectively, summing up to 128 neurons. Iterative optimization was employed to determine the optimal FNN structure, starting with a single hidden layer of 8 neurons and progressively experimenting with different configurations. Network parameters and the number of layers were tuned sequentially, maintaining a learning rate of 0.001, to achieve the best performance

using the training data.

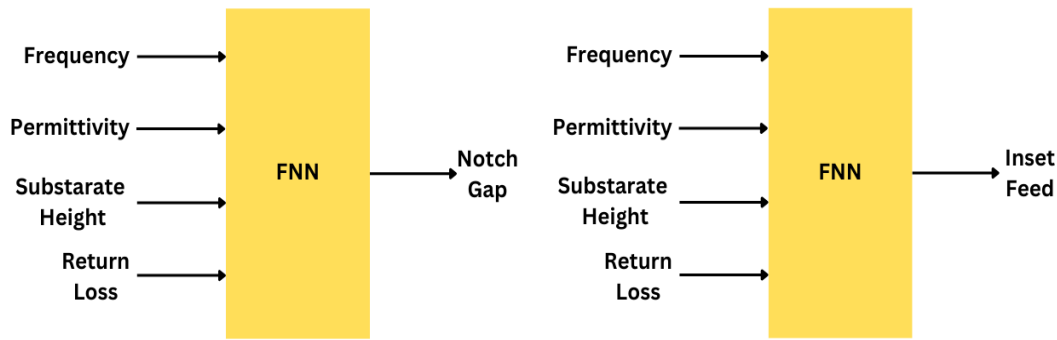


Fig. Schematic Diagram of Model

5.3 Grid Search CV

GridSearchCV is a tool used to enhance the performance of machine learning models by automating the process of hyperparameter tuning. It begins by defining the hyperparameters to test; the `param_grid` specifies different values for batch size, epochs, and optimizers that we want to experiment with. GridSearchCV then systematically tests every possible combination of these hyperparameters to determine which one performs the best. For each combination, it trains the model and evaluates its performance using cross-validation. After evaluating all combinations, GridSearchCV identifies the set of hyperparameters that results in the best model performance. In summary, GridSearchCV helps streamline the process of optimizing hyperparameters, thereby improving the accuracy and effectiveness of machine learning models.

5.4 Model Evaluation

Models Obtained for all parameter combination in GridSearchCV were then evaluated using test data to Identify the Best Model. Model Providing the best results was then saved to be used in application.

5.5 Calculated Values

Width of Antenna Patch(W) is calculated using :- c : Speed of Light

$$W = \frac{c}{2f_o \sqrt{\frac{(\epsilon_r + 1)}{2}}}$$

Length of Antenna Patch(L) is calculated using :

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{W} \right]^{-\frac{1}{2}}$$

$$L_{eff} = \frac{c}{2f_o \sqrt{\epsilon_{eff}}}$$

$$\Delta L = 0.412h \frac{(\epsilon_{eff} + 0.3) \left(\frac{W}{h} + 0.264 \right)}{(\epsilon_{eff} - 0.258) \left(\frac{W}{h} + 0.8 \right)}$$

$$L = L_{eff} - 2\Delta L$$

Width of 50Ω Impedance Line :

$$w = \frac{7.48 \times h}{e^{\left(Z_0 \frac{\sqrt{\epsilon_r + 1.41}}{87} \right)}} - 1.25 \times t$$

6. Application Development

6.1 Tools Used & UI

The entire user interface (UI) and backend framework of the application are built using Tkinter. Tkinter is utilized to create various GUI elements such as labels, buttons, and entry fields, as well as to handle user interactions and display results.

The application integrates a trained model to predict target values based on user-provided input data. This model, which was developed using machine learning techniques, is used to compute parameters like patch length, width, and feed line dimensions for microstrip antennas.

The application includes robust error handling. Specifically, the `'get_float_value()'` function is designed to ensure that only numeric values are entered into the input fields. If a non-numeric value is detected, it triggers an error message using Tkinter's message box to alert the user and prompt them to enter valid data.

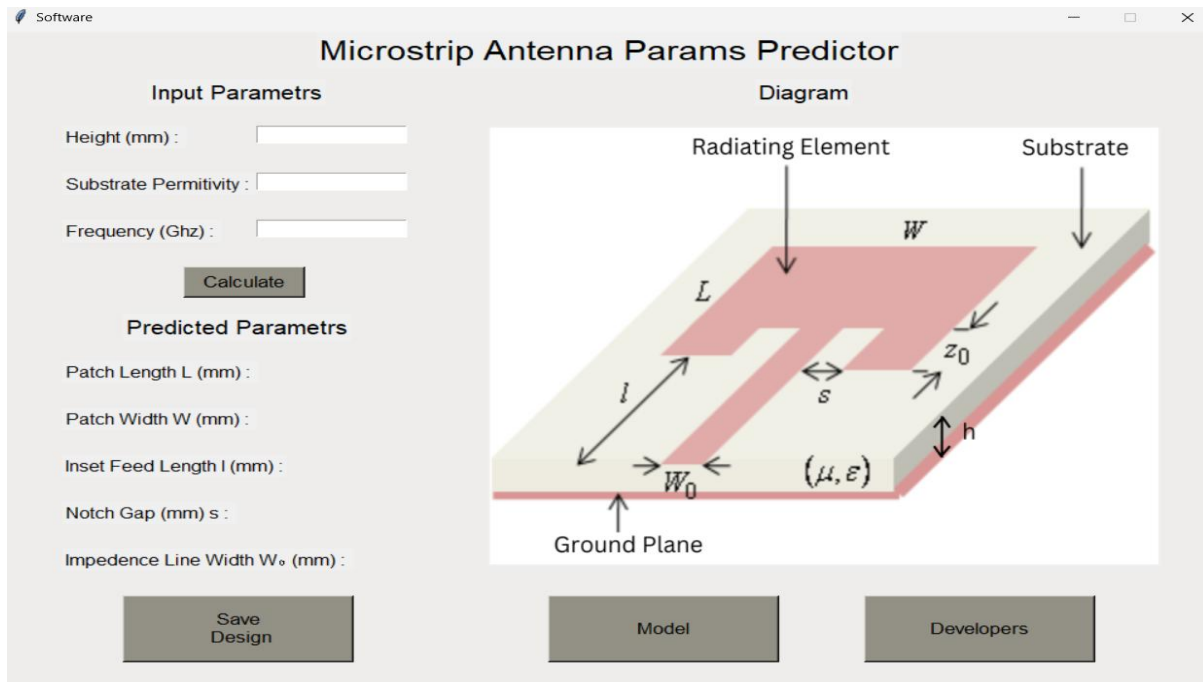


Fig. UI of Application

7. Results and Conclusion

7.1 Prediction

Height(mm)	Permittivity	Frequency(GHz)	Inset Feed Length(mm)			Notch Gap(mm)		
			Original Value	Predicted	% Error	Original Value	Predicted	% Error
1.6	2.2	0.915	26.5	26.43	-0.26 %	0.4	0.72	80 %
1.6	3.35	0.915	25.2	24.85	-1.38 %	0.6	0.58	-3.33 %
1.6	4.4	5.8	3.6	3.606	0.166 %	0.2	0.081	-59.5 %
0.8	2.2	2.45	9.8	9.84	0.408 %	0.4	0.27	-32.5 %
0.8	9.4	4	5	4.73	-5.4 %	0.2	0.80	300 %

7.2 Performance Metrics

- RMSE: Root Mean Squared Error. It's a commonly used metric to measure the accuracy of a model's predictions in regression tasks.
- MSE: Mean Squared Error is a common measure of the difference between the actual values and the values predicted by a model. It is calculated by taking the average of the squared differences between actual and predicted values.
- MAE: Mean Absolute Error is a metric used to measure the accuracy of a model in predicting continuous outcomes. It represents the average absolute difference between the predicted values and the actual values.
- The model's performance on the test dataset is evaluated with RMSE (0.7752), MSE (0.4549), and MAE (0.3175).

7.3 Conclusion

The proposed model efficiently predicts design parameters of Implantable Medical Patch Antennas (IMPA) using given values of frequency, dielectric constant of the substrate,

substrate height and return loss. The model successfully converges, demonstrating its effectiveness in predicting the design parameters for IMPA. Developed a UI to interact with model to predict targeted values.

8. Future Scope

To enhance the performance of your machine learning models, you can consider experimenting with advanced architectures and applying regularization techniques. For instance, more complex neural network architectures, such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), can be explored if they are suitable for your specific problem. Additionally, implementing regularization methods like dropout or L2 regularization can significantly improve model generalization and help reduce overfitting. By integrating these approaches, you can potentially achieve better performance and robustness in your models.

8.1 Similar Projects

1. Circular Microstrip Antenna Design Optimization:

- Objective: Predict and optimize parameters like radius, substrate height, and feed position to achieve desired performance metrics (e.g., resonance frequency, bandwidth).
- Data Inputs: Frequency, substrate permittivity, substrate height, antenna radius, and feed location.
- Models: Neural Networks (NN) to predict the optimal radius and feed position based on input features like frequency, substrate permittivity, and substrate height. GridSearchCV can be used to fine-tune hyperparameters.
- Outputs: Resonant frequency, bandwidth, and radiation pattern parameters.

2. Elliptical Microstrip Antenna:

- Objective: Estimate optimal dimensions (major and minor axes) and feed position for achieving specific design goals (e.g., gain, bandwidth).
- Data Inputs: Frequency, substrate permittivity, substrate height, major and minor axes of the ellipse, and feed position.
- Model: Random Forest Regressor, to estimate the major and minor axes of the elliptical antenna. The model inputs include frequency and substrate characteristics
- Outputs: Resonant frequency, gain, and radiation pattern characteristics.

3. Microstrip Antenna Failure Prediction:

- Objective: Predict potential failure modes or performance degradation based on design parameters and environmental conditions.
- Data Inputs: Design parameters, substrate materials, environmental conditions, and historical performance data.
- Model: Logistic Regression to predict the likelihood of failure or performance degradation based on design parameters, environmental conditions, and historical performance data.
- Outputs: Predicted likelihood of failure or performance issues.

9. Citations

- [1]. [Artificial Neural Network Model to Predict the Design Parameters of Inset-fed Microstrip Patch Antenna.](#)
- [2]. [An Intelligent Antenna Synthesis Method Based on Machine Learning.](#)
- [3]. [Design Formula for Inset Fed Microstrip Patch Antenna.](#)
- [4]. [Tensorflow Documentation](#)
- [5]. [Pandas Documentation](#)
- [6]. [Tkinter Documentation](#)